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Intergalactic Interactions - Network Biology in Rodents from Spaceflight and Terra Firma

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Abstract

This study, utilizing data from the NASA Gene Lab Data Repository, investigates the impact of spaceflight on rodents, analyzing changes in lung, colorectal, and skin tissues. We examined RNA-seq data from rodents that had experienced space flight, comparing them with control groups that remained on Earth. Our focus was to better decipher the molecular alterations induced by the unique conditions of space. A comprehensive initial cohort of rodents was selected, ensuring the integrity and completeness of genomic data. We employed Proteinarium, a sophisticated multi-sample analysis tool, to explore and compare protein-protein interaction (PPI) networks between space flight-exposed and control rodents. This approach allowed us to identify significant clusters and distinct PPI networks in the space flight group, which were not present in the control group. Our results revealed that space flight leads to unique molecular changes, with certain genes becoming upregulated in response to the extraterrestrial environment. These changes were tissue-specific, indicating varied physiological responses across different organ systems. We also identified distinct gene hub targets for drug repurposing, which could be evaluated in future studies. The present study contributes to a deeper understanding of how spaceflight affects living organisms at the molecular level, providing insights that could inform future space missions and enhance our knowledge of biological systems in spaceflight environments.

Introduction

In the vast expanse of space, the microscopic intricacies of biological systems unfold in ways yet to be fully understood. Rodent Research presents an unparalleled opportunity to delve deeper into these mysteries (https://www.nasa.gov/general/rodent-research/). This study focuses on the impact of spaceflight conditions on network biology in rodents, providing invaluable insights into how biology adapts at the cellular and molecular levels in extraterrestrial environments. By juxtaposing data from spaceflight and earthbound conditions, 'Terra Firma' - this paper sheds light on the intergalactic interactions that shape biological systems during spaceflight. Network biology-based approaches not only enhance our comprehension of life in space but also possess the potential to revolutionize our understanding of terrestrial biological systems by linking phenotypic differences with related, over-expressed genes.

In the realm of genetic exploration, the nuanced interplay of genes and their role in complex diseases has been a focus of relentless curiosity. Genome-wide association studies (GWAS)[1] have emerged as a pivotal approach in the quest to understand these intricacies, along with any variants associated with a particular finding in relation to space flight. One inherent limitation of GWAS in focusing primarily on the Common Disease, Common Variant Hypothesis[2], which proposes that common diseases arise from the additive effects of variants in individual genes. But the multifaceted nature of genetic diseases suggests a richer tapestry of interactions.

Newer perspectives posit that rather than individual genes, specific networks and pathways might be the linchpin, with groups of patients sharing variants that lead to similar phenotypic expressions. In some

cases, different networks and pathways might culminate in the same phenotypic outcome, delineating varied patient subgroups. Within these intricate networks, certain nodes or junctions may have a heightened significance, potentially due to natural selection processes which might lead to pathogenic variants clustering within specific pathways or areas of high interaction.

Navigating this intricate web of interactions, protein-protein interaction (PPI) networks have come to the forefront[3]. Advanced screening techniques have enabled the construction of detailed PPI networks, serving as dynamic blueprints of how genes functionally interplay. These networks aren't merely abstract maps; they reveal that genes with similar phenotypic outcomes often physically interact at the protein level[4]. This is evident not just in model organisms, but also in humans. Proteins with analogous disease phenotypes not only interact but also tend to gravitate towards the same network domains. PPI networks have proved instrumental, particularly in shedding light on elusive genes and subnetworks associated with intricate diseases.

Amid this backdrop, we aim to apply Proteinarium[5], a cutting-edge protein-protein interaction tool. Crafted to penetrate the depths of complex diseases and phenotypes, Proteinarium seeks to highlight clusters of patients unified by shared networks. Beyond mere identification, this tool bolsters the analysis of experimental data by pinpointing disease-centric biological networks and mapping patient clusters.

Materials and Methods

Study Design

Our study design utilized datasets from the NASA Gene Lab data repository for spaceflight experiments[2]: https://osdr.nasa.gov/bio/repo/search?q=&data_source=cgene,alsda&data_type=study. We focused specifically on the effects of space flight on rodents. Accessing RNA-seq and genomic data of rodents that had been part of space missions, aiming to understand the molecular changes induced by space environments.

The initial cohort included a significant number of rodent specimens, carefully chosen based on the availability and completeness of their genomic data. We prioritized datasets that provided a comprehensive view of the molecular alterations occurring in lung, colorectal, and skin tissues of these rodents. The selection process involved excluding cases with incomplete or non-relevant genomic information, ensuring that our final cohort comprised specimens with high-quality, relevant data for our research objectives.

By leveraging the NASA Gene Lab's extensive repository, our study aimed to uncover novel insights into the biological impact of space flight, contributing to a deeper understanding of the physiological and molecular adaptations in space-exposed rodents.

Data Processing

We used R programming language in order to create scripts that selected the 100 most optimal seed genes from the Basal, Ground and Vivarium Control groups, as well as the Flight Groups of lung, colorectal, and skin tissues. These genes are selected based on their frequency of expression in the organism from all datasets using an R script with the libraries readr, dplyr, tidyr, stringr, DataCombine, biomaRt, DECIPHER, dendextend, BiocManager, and corrplot. The script uses loops in order to create patient ID matrices and places the patients in different matrices according to their groups (Flight, Ground Control, Vivarium Control, Case Control, Basal Control) while omitting the null values. Using ID conversion functions, gene names were converted into human Z-Scores from rat ensembl Gene IDs. 100 genes with the highest Z-Score values were then selected and exported as Comma-Separated Values (.csv) files. Then, Excel functions are utilized to format the exports and convert them into Text (.txt) files. These files were uploaded into Atom Text Editor for the last formatting, then exported as clean data for processing.

Proteinarium

In our study, we employed Proteinarium[5], a sophisticated multi-sample analysis tool, to explore PPI networks in rats subjected to space flight and their control counterparts. This tool was instrumental in identifying clusters of rats with shared PPI networks, providing insights into the molecular impact of space flight on lung, colorectal, and skin tissues.

Proteinarium utilizes experimentally validated PPI data from the STRING database to create network graphs for each rat, based on a set of predetermined seed genes. The tool calculates the similarity between each pair of network graphs using the Jaccard distance. This similarity data is then compiled into a matrix, serving as the basis for clustering the rats' network graphs. The clustering method employed was the Unweighted Pair Group Method with Arithmetic Mean (UPGMA), which organizes rats based on the similarity of their network graphs.

The significance of each cluster was determined using Fisher's exact tests, considering the distribution of space flight and control rats. The network graphs were constructed with a minimum path length of 2, focusing on direct interactions or those mediated by a single intermediary protein. We selected 100 seed genes as optimal, as suggested by Proteinarium developers[5].

Finally, Proteinarium compared the PPI networks of these selected control rats with those of the space flight-exposed rats. The analysis produced a dendrogram and identified significant clusters, offering valuable insights for subsequent investigations into the molecular effects of space flight on these rodents.

Results

Our study, drawing from the NASA Gene Lab data repository, analyzed the effects of space flight on rodent specimens, focusing on lung, colorectal, and skin tissues. We identified 19, 40 and 5 space flight rodents and 35, 57 and 17 control rodents for colorectal (GLDS-247)[6], dorsal skin (GLDS 238[7], 240[8], 243[9]), and lung (GLDS-248)[10] groups respectively. All organ system showed distinct networks with p

values less than 0.05 in their relative comparisons. *HTR6, GNRHR, PROK1, and NPHS2* were the most uniquely upregulated genes among the colorectal data in the space flight-exposed rodents, while for dorsal skin data they were *APOA2, AFP, PCP2*, and *CRP* genes and for lung they were *DERL3, NMBR, TRIM69, MAP3K14*, and *SYNV1*. Most of these genes are associated with certain diseases, deficiencies, and types of cancers, such as ovarian cancer, gastric cancer, and pancreatic duct cancer.

Space Flight and Control Cohorts had Distinct PPI Networks

Proteinarium Dendrograms revealed significant clusters from space flight and control rodents. Numerous clusters were chosen for further investigation based on the degree of domination according to the exposure to space flight. Significant clusters dominated by space flight included upregulated genes unique to rodents exposed to space flight and not seen in control rodents.

Colorectal:

The hub or most connected proteins exclusive to the Flight group networks include *HTR6, GNRHR, PROK1*, and *NPHS2*, with *HTR6* being the most highly connected protein in the network.

HTR6 belongs to the seven-transmembrane G protein-coupled receptor family of proteins, and critically links to AMP-dependent signaling pathway and cholinergic neural transmission in the brain. Accumulating evidence in both rodent and human studies suggests that pharmacological blockade of *HTR6* signaling improves memory performance[11–16] and mitigates cognitive deficits in animal models of schizophrenia[17]. Several antidepressants and antipsychotic drugs have a high affinity for this receptor, including but not limited to Asenapine, Clozapine, Iloperidone, Imipramine, Amoxapine, Doxepin, Amitriptyline, Mianserin, Zotepine, Haloperidol, and Sertindole. Interestingly, the study of this neurotransmitter receptor in human breast cancer has revealed a significant correlation with patient outcomes and uncovered *HTR6* as a gene associated with survival[18].

GNRHR encodes for the type 1 gonadotropin-releasing hormone receptor and is expressed on the surface of pituitary gonadotrope cells as well as lymphocytes, breast, ovary, and prostate. Defects in this gene are a cause of hypogonadotropic hypogonadism (HH), and overexpression might result in ovarian cancer and other hormone-related cancers[19]. This gene can be co-expressed with *GNRHR-II* to decrease its cell surface expression using methods of alternative splicing[20]. Multiple GnRH antagonists, such as Cetrorelix, Ganirelix, and Teverelix, in inhibiting GnRH-mediated intracellular signaling, target *GNRHR*[21].

PROK1 codes for Prokineticin-1 protein that induces proliferation, migration, and fenestration in capillary endothelial cells derived from endocrine glands. *PROK1* expression correlates with malignancy in colorectal cancer along with prostate cancer, neuroblastoma, thyroid cancer, pancreatic duct cancer, gastric cancer, and small intestinal cancer, indicating that *PROK1* is important for malignant tumor formation[22–26]. *RCAN1-4* gene down modulates calcineurin/NFAT pathway, and can play a regulatory role for the inhibition of *PROK1*-induced IL-8 expression[27]. Triazinedione Derivatives can be used as Prok1 receptor antagonists[28].

The *NPHS2* gene encodes for podocin. Podocin is a protein primarily found in podocytes, which are located in glomeruli. Podocin likely helps bring other proteins that are needed for a functional slit diaphragm to the podocyte cell surface, as well as cell signaling.*NPHS2* is also related to Type 2 Nephrotic Syndrome. Corticosteroids (prednisone), cyclophosphamide, and cyclosporine are drugs usually used against NPHS-caused Nephrotic Syndrome.

Dorsal Skin:

The hub or most connected proteins exclusive to the Flight group networks include *APOA2, AFP, PCP2*, and *CRP* with *APOA2* being the most highly connected protein in the network.

APOA2 is the second most abundant protein of the high-density lipoprotein particles. This is associated with pathways such as the assembly, remodeling, and clearance of plasma lipoproteins, as well as with Familial Hyperlipidemia Type 1. It is closely related to lipid binding. Defects in this gene may result in apolipoprotein A-II deficiency or hypercholesterolemia, while overexpression is associated with obesity and insulin resistance[29]. In the liver, overexpression of the human*FOXA3* gene, which encodes a member of the forkhead class of DNA-binding proteins, significantly induced several genes involved in HDL biogenesis, including *APOA2*[30]. These hepatocyte nuclear factors function as transcriptional activators for liver-specific transcripts such as albumin and transthyretin, while also interacting with chromatin. APO mimetic peptides are designed to mimic structural and functional features of APOs, which show the therapeutic potential for cancer[31].

The AFP gene is responsible for producing alpha-fetoprotein, a significant protein found in plasma that is synthesized by the yolk sac and liver during the developmental stages of a fetus. Overexpression of *AFP* could enhance the proliferation, invasion, and migration of hepatocellular carcinoma (HCC) cells. *AFP*-derived peptides[32], AFP fragments, and recombinant AFP (AFP-inhibiting fragments, AIFs) can inhibit the binding of AFP to intracellular proteins for the treatment of HCC and other cancers[33]. This strategy for treating hepatocellular carcinoma (HCC) and other cancers revolves around using peptides derived from alpha-fetoprotein (AFP), fragments of AFP, and recombinant AFP. These components, known as AFP-inhibiting fragments (AIFs), are employed to prevent AFP from binding to its receptors or to proteins within cells.

PCP2 is a protein-coding gene that is related to pathways such as GPCR downstream signaling and Signal Transduction. Diseases associated with *PCP2* include Neuropathy[34], and Gaucher Disease[35]. Pertussis toxin treatment can completely block Pcp2-mediated increased neurite growth[36]. Transient transfection of the β-adrenergic receptor kinase C-terminus to prevent signaling through Gβγ in Pcp2-PC12 cells also inhibits the Pcp2-induced phenotype and reduces the Pcp2-stimulated Ras activation.

CRP encodes C-reactive Protein, which belongs to the pentraxin family involved in complement activation and amplification via communication with complement initiation pattern recognition molecules. Overexpression of *CRP* may have a pathogenic role in the development of pressure overload-induced cardiac remodeling, possibly through enhanced inflammation and oxidative stress[37].

Promising new research suggests that treating rats with a rat CRP-specific ASO causes a reduction of blood CRP[38]. This was found to be associated with improved heart function and pathology after myocardial infarction.

Lung:

The hub or most connected proteins exclusive to the Flight group networks include *DERL3, NMBR, TRIM69, MAP3K14*, and *SYNV1* with *DERL3* being one of the most highly connected proteins in the network.

DERL3 encodes Derlin-3 protein that belongs to the derlin family and resides in the endoplasmic reticulum to degrade misfolded glycoproteins. Overexpression of *DERL* 3 is associated with malignant phenotype of breast cancer cells[39].*DERL3* can be repressed with DNA methylation techniques[40].The results from past research indicated that the *DERL3* low-expression group significantly responded sensitively to the chemotherapeutic agents gemcitabine, etoposide, and docetaxel[41].

NMBR encodes a 7-transmembrane G protein-coupled receptor that binds neuromedin B, which is a growth factor and mitogen for gastrointestinal epithelial tissue[34] and normal and neoplastic lung. It is also associated with smooth muscle contraction, neuronal responses, and the regulation of cell growth. Diseases associated with *NMBR* include osteosarcoma[38]. Research found that *NMBR* can be inhibited using CRISPR-Cpf1-based transcription factors[42]. The C terminus of *NMBR* was fused with a tail sequence from *AVPR2* and exhibited little basal activity without the addition of ligands. It is also shown that an NMB-R antagonist, PD168368, suppresses migration and invasion of the human breast cancer cell line MDA-MB-231[43].

The *TRIM69* gene codes for a protein belonging to the RING-B-box-coiled-coil (RBCC) family. Its mouse equivalent is expressed exclusively in germ cells during the round spermatid stages of spermatogenesis. When overexpressed, this gene can trigger apoptosis.

MAP3K14 encodes mitogen-activated protein kinase kinasekinase 14. This kinase binds to TRAF2 and stimulates NF-kappaB activity. The protein participates in an NF-kappaB-inducing signaling cascade common to receptors of the tumor-necrosis/nerve-growth factor family.

A study showed that mice with reduced *MAP3K14* activity showed less kidney dysfunction, inflammation, and apoptosis[44]. Exosome-mediated delivery of gaa-miR-20a-5p has been shown to target *MAP3K14*[45].

SYNV1 encodes a protein involved in endoplasmic reticulum (ER)-associated degradation. The protein encoded by this gene functions to eliminate unfolded proteins, which accumulate during ER stress, by transporting them in a retrograde manner from the ER to the cytosol. This protein also uses the ubiquitin-proteasome system for additional degradation of unfolded proteins. Mice overexpressing *SYNV1* develop severe spontaneous arthropathy[46]. According to previous research, cryptochlorogenic acid (CCA) shows strong affinity with the binding pocket of *SEL1L* and inhibits SEL1L protein coded by *SYNV1*[47].

Discussion

Our project evaluated gene expression in rats exposed to spatial conditions, such as microgravity and radiation, which can only partly be simulated on Earth due to the complexity[48, 49]. We observed that some genes are uniquely expressed in spaceflight rodents, while others are not expressed or have reduced expression, unlike in the group of rats on Earth. Our project along with many future projects, and new methods proposed is a study that has the potential to be reflected in not only human tissues, but also other organisms. A gene analysis project about transcriptomic datasets of plant model organism Arabidopsis thaliana[50] is a similar work that has been conducted in the past. A newly proposed method for analysis of human tissues is human tissuechips[51], which are the next step in transferring projects conducted on animals to human cells. In addition to future research at the International Space Station (ISS), new platforms such as CubeSats, SmallSat, and Lunar Orbital Gateway[51], are possible future grounds for astrobiological and astrochemical research.

The use of Proteinarium[5] for analyzing PPI networks in rodents subjected to space flight provides a novel approach for identifying gene targets in drug repurposing. This method, drawing on data from NASA's Gene Lab[2], has uncovered unique molecular changes in spaceflight-exposed rodents, specifically in lung, colorectal, and skin tissues. The study's significant discovery of distinct gene hubs, like *HTR6* in colorectal tissues and *APOA2* in skin tissues, offers new insights into the physiological impact of space conditions. These findings are crucial for understanding the complex interplay of genes in extreme environments and could guide the repurposing of existing drugs to target similar molecular pathways in human diseases.

After the analysis of the colon, dorsal skin, and lung datasets, *HTR6, APOA2, and DERL3* genes are determined to be the most uniquely upregulated in the space flight-exposed rodents. Asenapine, Clozapine, Iloperidone, Imipramine, Amoxapine, Doxepin, Amitriptyline, Mianserin, Zotepine, Haloperidol, and Sertindole are drugs known to target *HTR6*; APO mimetic peptides were proposed as a future study topic for *APOA2*; gemcitabine, etoposide, and docetaxel are proposed inhibitory drugs for the protein encoded by *DERL3*.

It is important to pitfalls in the present study. The obvious is the small sample sizes for rats in our study. As well we combined different controls and flight cohorts to increase our sample size which was not ideal. It is also worth mentioning that previous studies examining physiological changes in mice sent to the ISS found challenges associated with sample collection in space and the confounding factors it introduces[52]. This includes changes in the transcriptome due to preservation methods, which can overshadow differences between spaceflight and ground samples, suggesting the need for careful consideration of these factors in future space research.

Our initiative is bolstered by the recent introduction of the GeneLab Application Programming Interface (API), which is tailor-made to enhance tools used for visualizing processed omics data[53]. GeneLab's commitment to utilizing space flight data housed in its repository is pivotal for sparking new discoveries and formulating fresh hypotheses. Looking ahead, extending research to include human tissue, coupled

with the integration of medical imaging including whole slide images, holds promise for comprehensive multi-omics analysis, and a deeper understanding of the alterations caused by space flight.

Conclusions

This study sheds light on lung, colorectal, and skin cancers in rodents, focusing on molecular subtypes that are distinct and not involving control rodents. Our research identified unique PPI networks with a central role played by different hub genes upregulated by space flight. These networks could have varied clinical implications for space flight, which are independent of the conventional molecular genotype. The insights gained here are critical for better understanding the molecular characteristics of space flight. Future work can help provide a foundation for developing targeted therapeutic strategies, better tailored to the specific network biology subtypes associated with spaceflight.

Declarations

Data Availability Statement

All data collected and analyzed in this study are available at https://osdr.nasa.gov/bio/repo/search?q=&data_source=cgene,alsda&data_type=study.

Conflicts of Interest

Author S.H. is the founder and has ownership equity in Odyssey HealthCare Solutions, Inc. The remaining authors declare no conflicts of interest.

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Figures

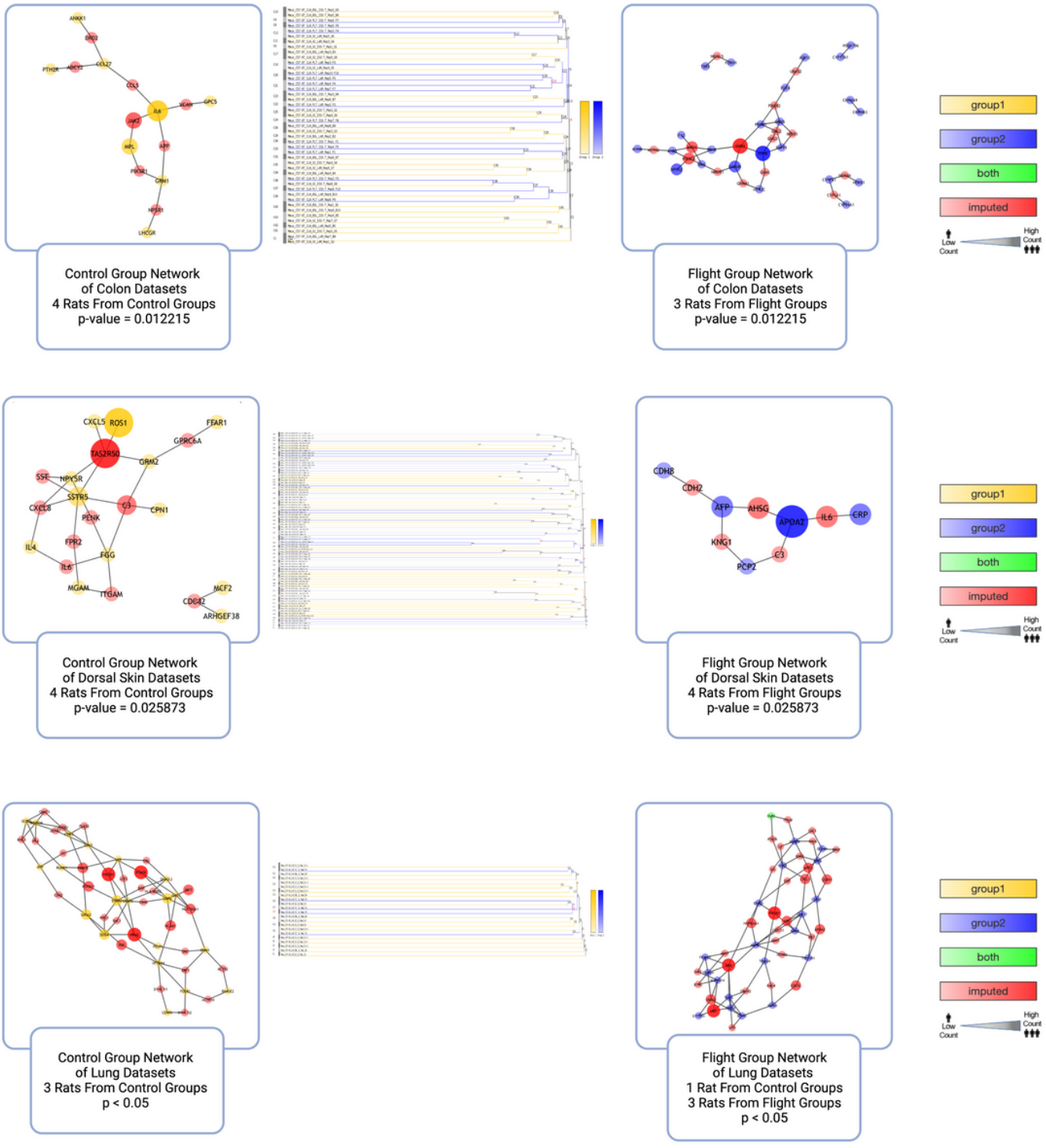


Figure 1

Dendrograms and consensus networks were generated to visualize the distinct molecular portraits of space flight-exposed rodents compared to control rodents. In these visual representations, larger nodes indicate a higher level of connectivity within the networks. Rodents from the space flight cohort are represented with blue branches, contributing to the space flight-specific consensus network on the right side of the diagram. Conversely, the control rodents are depicted with orange branches, forming the consensus network for the control group on the left. In these networks, blue nodes represent genes that

are uniquely upregulated in space flight-exposed rodents, while yellow nodes are genes unique to the control group. Green nodes indicate genes that are present in both space flight and control rodent networks within the same cluster. Finally, red nodes denote imputed genes, which are used to infer potential interactions or pathways. Figures were created using Biorender (https://www.biorender.com/).